

# ML Scan Parameters

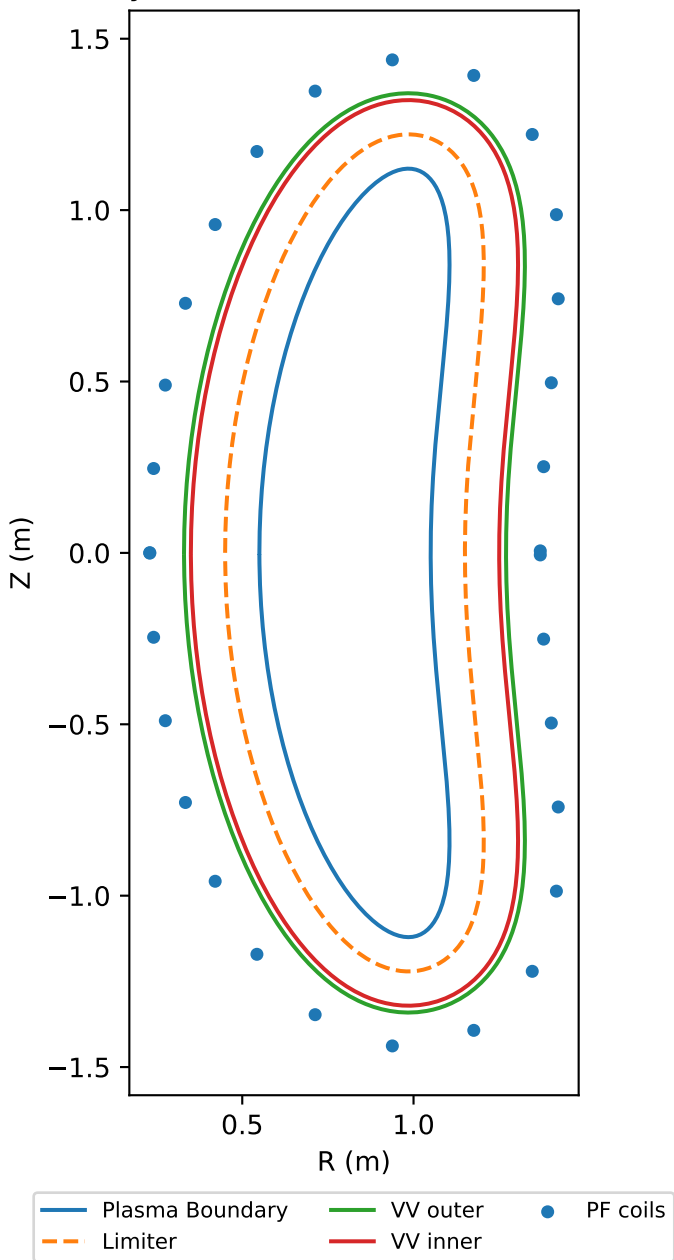
C-shape plasma  $(r,z) = (A + B\sin t + C\cos 2t, F + G\cos t - H\sin 2t)$

A=0.9200 B=0.2500 C=0.1200  
G=1.1000 H=0.1100 F=0.0  
R0=0.8000 m kappa\_target=4.400 delta=-0.110  
B0=1.8 T Ip=1.00 MA

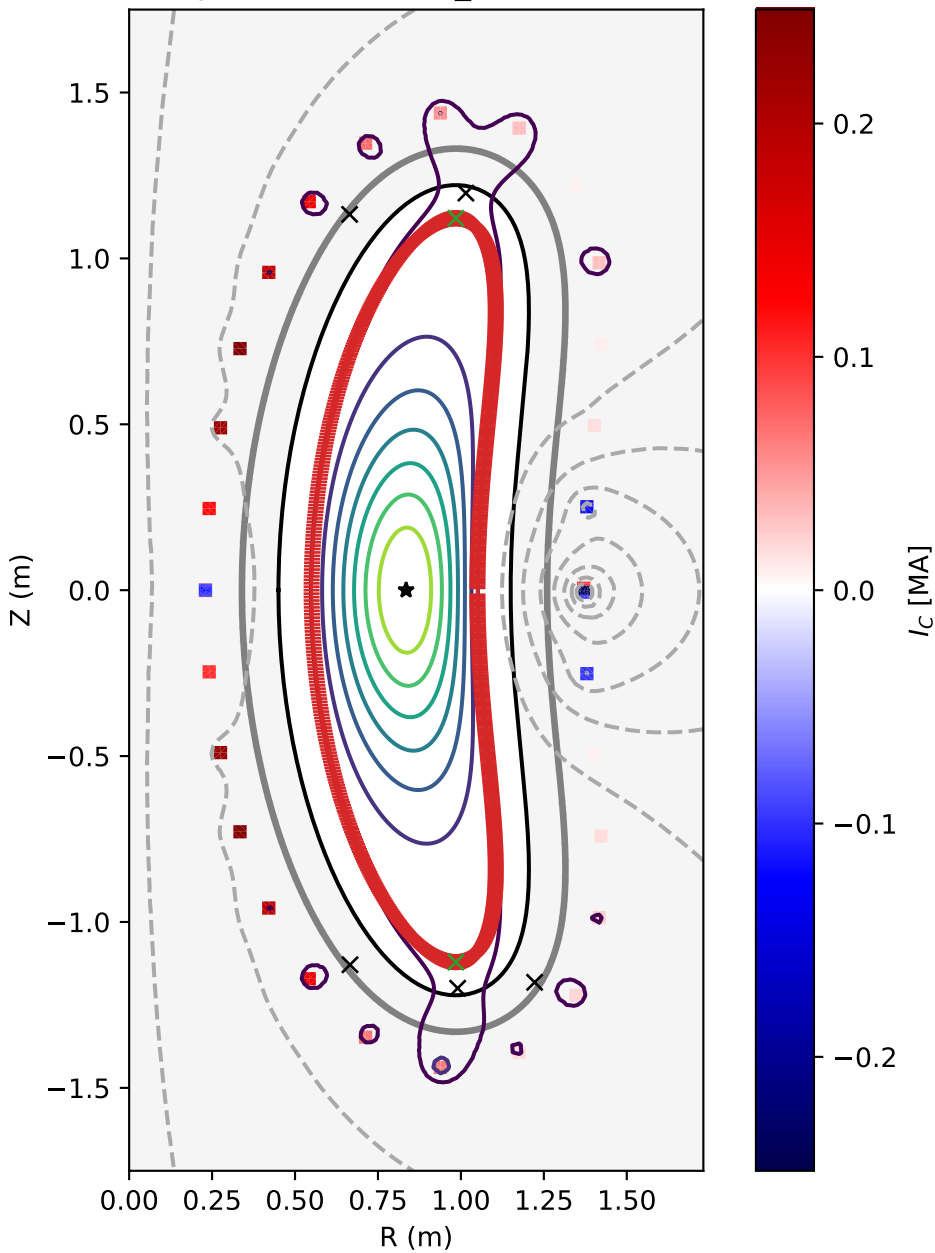
Plasma profiles:  
pp\_prof:  $(1 - \psi^{0.738})^{1.519}$   
ffp\_prof:  $(1 - \psi^{0.859})^{0.734}$

Scan params: B=2.5 C=12 G=11 H=11  
alpha\_pp=0.738 gamma\_pp=1.519  
alpha\_ff=0.859 gamma\_ff=0.734

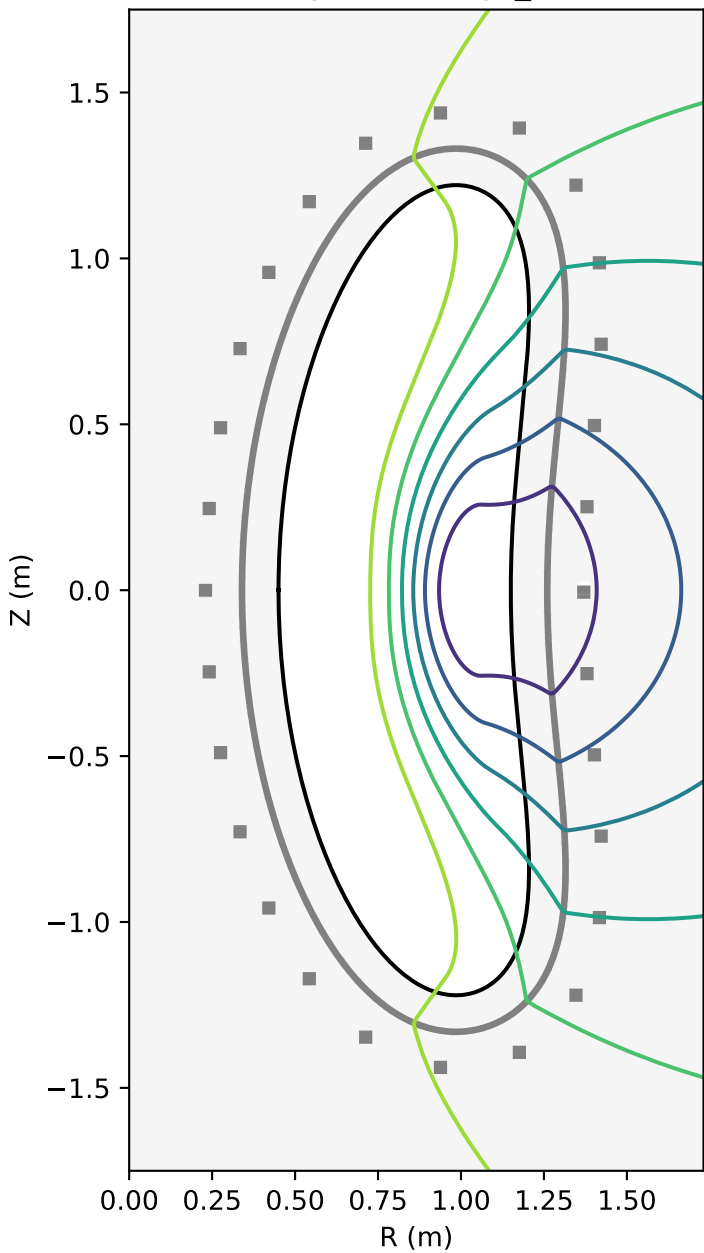
Geometry B=2.5, C=12, G=11, H=11,  $\kappa=4.4$



Equilibrium flux  $\kappa_{\text{actual}}=4.01$



Unstable eigenmode  $\gamma\tau_w = -1.000$



# Vertical Stability Analysis

## Vertical Stability Results

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B=2.5 C=12 G=11 H=11

alpha\_pp=0.738 gamma\_pp=1.519

alpha\_ff=0.859 gamma\_ff=0.734

kappa\_target = 4.400

kappa\_actual = 4.013

beta\_N = 0.454 %

growth rate (gamma) = -4.6620E+01 s-1

wall time (tau\_w) = 2.1450E-02 s

gamma \* tau\_w = -1.0000

max coil current = 0.249 MA

Category: passive stable

axis\_offset\_pct = 4.35%

Equilibrium Statistics

|                         |                                 |
|-------------------------|---------------------------------|
| Equilibrium Statistics: |                                 |
| Topology                | = Limited                       |
| Toroidal Current [A]    | = 1.0000E+06                    |
| Current Centroid [m]    | = 0.831 0.000                   |
| Magnetic Axis [m]       | = 0.835 0.000                   |
| Elongation              | = 4.013 (U: 4.012, L: 4.014)    |
| Triangularity           | = -0.560 (U: -0.564, L: -0.555) |
| Plasma Volume [m^3]     | = 5.468                         |
| q_0, q_95               | = 0.998 9.578                   |
| Peak Pressure [Pa]      | = 1.0654E+05                    |
| Stored Energy [J]       | = 9.3509E+04                    |
| <Beta_pol> [%]          | = 42.1717                       |
| <Beta_tor> [%]          | = 0.9436                        |
| <Beta_n> [%]            | = 0.4536                        |
| Diamagnetic flux [Wb]   | = 1.0256E-02                    |
| Toroidal flux [Wb]      | = 1.7547E+00                    |
| l_i                     | = 1.0300                        |